Vismay Vyas

(630) 808-2228 · vismayvyas392@gmail.com · www.linkedin.com/in/vismay-vyas

Education

Purdue University, West Lafayette, IN — Expected: May 2027
Master of Science in Mechanical Engineering — GPA 3.80/4.00

University of Illinois Urbana-Champaign, Champaign, IL — December 2024
Bachelor of Science in Mechanical Engineering, 3x Dean's List — GPA 3.74/4.00

Work Experience

Purdue University, West Lafayette, IN — Jan 2025 – Present
Graduate Research Assistant

- Devised a novel thermal energy storage direct-integration method into a vapor compression cycle with staged expansion processes to leverage an intermediate system pressure for effective storage utilization
  - Requires smaller CapEx and OpEx while maintaining actively controllable energy storage
- Created a modular DAE approach to auto-generate dynamic system models and sweep enumerated thermal management system architectures for transient open-loop simulation of F18 aircraft
  - Combats thermal system design fixation by searching a wider space of candidate architectures

University of Illinois at Urbana-Champaign, Champaign, IL — August 2024 – December 2024
Undergraduate Researcher

- Used physics-informed neural networks to approximate solutions to nonlinear partial differential equations for analysis of annular, multiphase flow in micro-channel heat exchangers

Intelsat LLC, Chicago, IL — May 2024 – August 2024
Aircraft Technical Services, Mechanical Engineering Intern

- Conducted structural analysis on plane parts using emergency landing loads to meet FAA regulations

Atkore, Hobart, IN — January 2024 – May 2024
Mechanical Engineering Consultant

- Developed end-to-end pipe fitting for variable material conduits (UL standards 651 & 157)
  - Designed grooved elastomeric sleeve for compressive grip and waterproofed sealing
  - Performed Finite Element Analaysis to simulate UL tests and determine safety factors

Patents and Publications

- [DUAL-EXPANSION VALVE ARCHITECTURE IN TES-INTEGRATED VAPOR COMPRESSION SYSTEMS] ([**V. Vyas**] et al.), Application No. [64/106,792], Filed [July 2026] (Patent Pending)
- **V. Vyas** et al., "Graph-Based Dynamic Modeling of a PAO Loop for Aircraft Thermal Management Systems," in AIAA SCITECH 2026.
- J. Pisani, C. Aksland, P. Renkert, J. Broniszewski, **V. Vyas**, et al., "Graph-Based Modeling, Control, and Optimization for Multi-Domain and Multi-Timescale Energy Systems," in ACC 2026.
- **V. Vyas** et al., "A Dual-Expansion Valve Architecture for Intermediate Pressure Control in TES-Integrated Vapor Compression Systems," in Herrick Conferences 2026.

Technical Skills

MATLAB, Simulink, Python, Finite Element Analysis (Abaqus, Femap with NX Nastran), Computer-Aided Design (SolidWorks, Autodesk Fusion 360, Creo Parametric)